

Pulsation of a Subdwarf B Star

ASTR 302, SPRING 2026

Zach Haarz, Lauren Jones, Thomas Nguyen, Ava Anderson



Introduction

Through observing the light curve and periodogram of a subdwarf B star, we will be able to mold the data into a format that allows us to derive the pulsation frequencies we observe. The short-period of our target star makes it an attractive candidate for being able to get a large number of pulsation periods over just one night of observing. Through this program, we observe the p-modes of a subdwarf B star to further understand the asteroseismological processes of the star through comparison with existing test data and other stars within the field of observation.

What is a subdwarf B star?

- Burn helium at their core
- Blue end of the horizontal branch

Kappa opacity mechanism

- as opacity increases with temperature (opposite to what occurs in normal, non-pulsating stars) the star becomes unstable pushing it to where it will then pulsate causing the period we are able to observe
- Areas where pulsations are excited are called instability strips

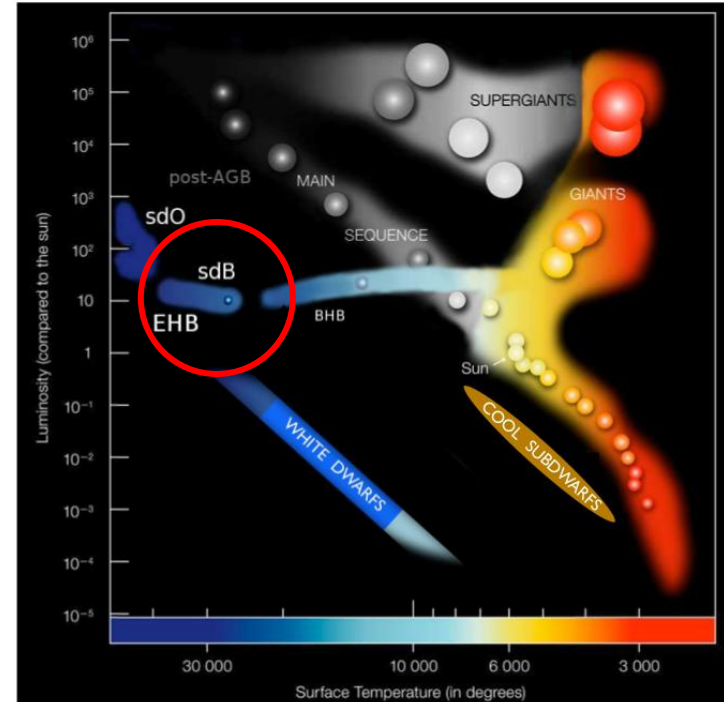


Image Credit: Annual Review Astronomy & Astrophysics.

Target Information

Object Name: 2M0753+2324

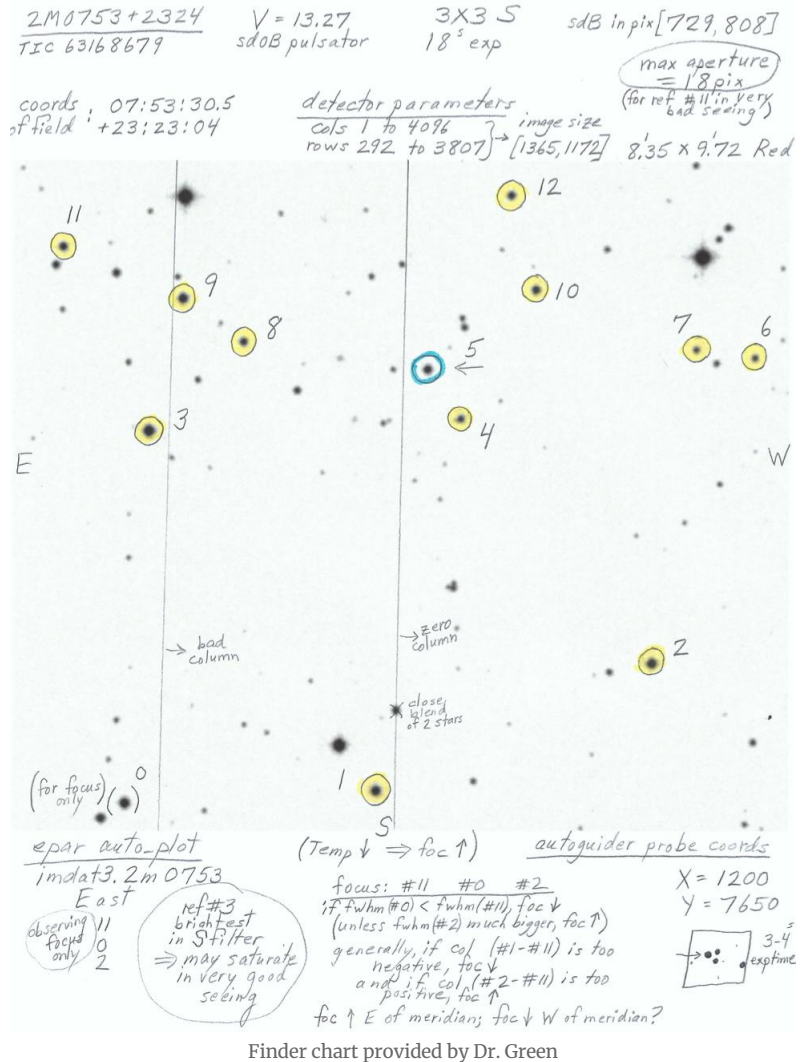
RA and Dec: 07:53:29.1 +23:24:36

V Magnitude: 13.27

Object Type: Pulsating sdB star

Viewing Time: March 6-7th ~7pm-11pm (cut short by cloud coverage)

Telescope: The Steward
Observatory 61" Kuiper Telescope
Mt. Bigelow, Tucson, AZ



Methods/Observations

→ By studying how brightness changes over time, more specifically within the pulsation periods of the star we observed, we use the data gathered to make a periodogram.

Observing the light curve allows us to analyze p-modes of the star giving us more information on the stellar structure and/or asteroseismology involving this star.

Steps to reduce data and obtain light curve

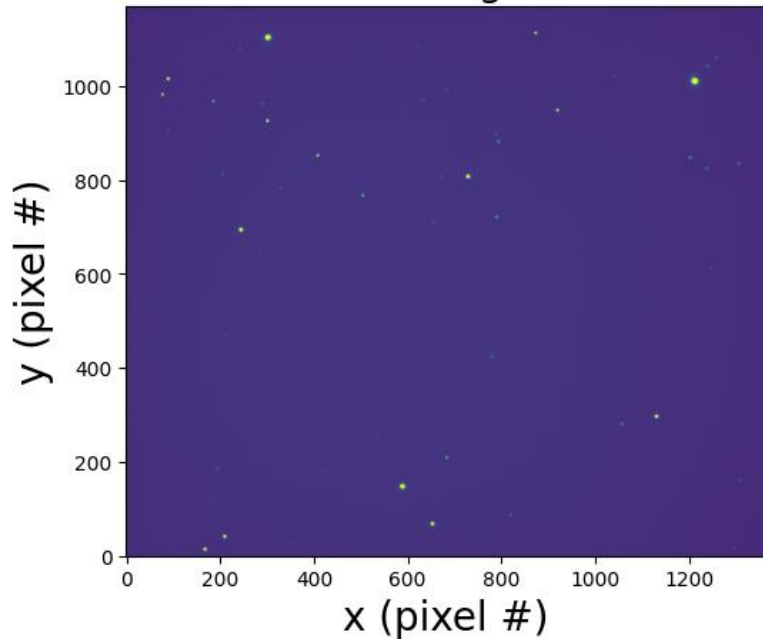
1. Image reduction through flat fielding and bias subtraction
2. Cosmic ray removal (proved unnecessary through recommendation of Dr. Green)

NOTE: always a good precaution to take

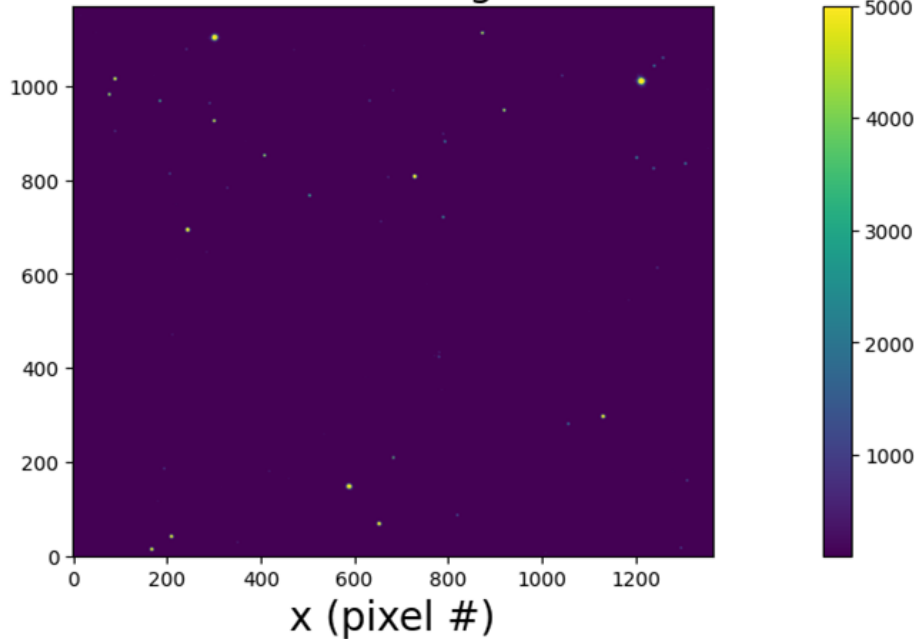
1. Photometric data reduction
2. Obtain Lomb-Scargle periodogram

Pre- and Post-Reduction Starfields

Raw Science Image of 2M0753



Reduced Science Image of 2M0753



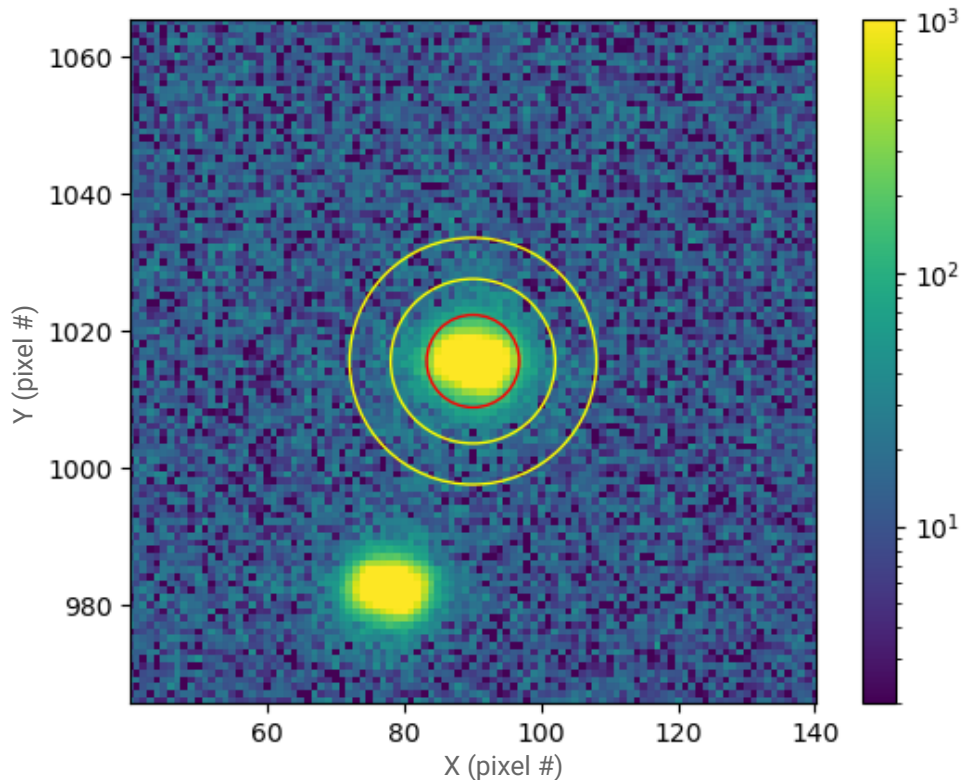
Data Analysis and Photometry

High-brightness pulsating variable target star,
large number of dispersed reference stars,
atmospheric noise: relative photometry

Co-wrote 3 pipelines for cross-consistency and
implementation of different techniques:

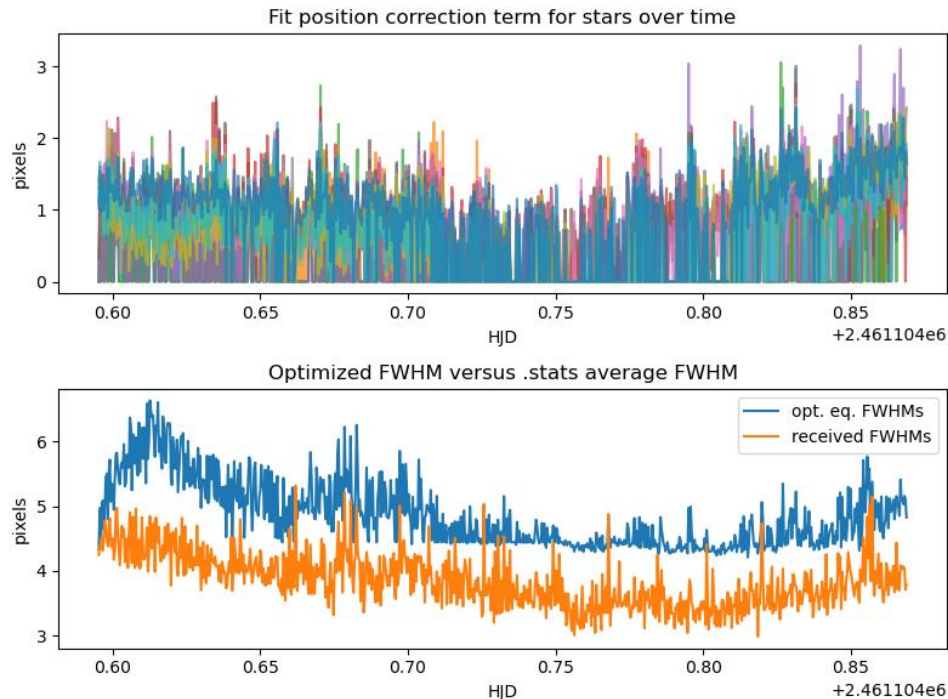
- Heuristic or optimized aperture parameters?
- Sources of noise?
- How should we extract frequencies?

Final result is Lomb-Scargle (mildly irregular
observations) periodogram of normalized
relative stellar flux

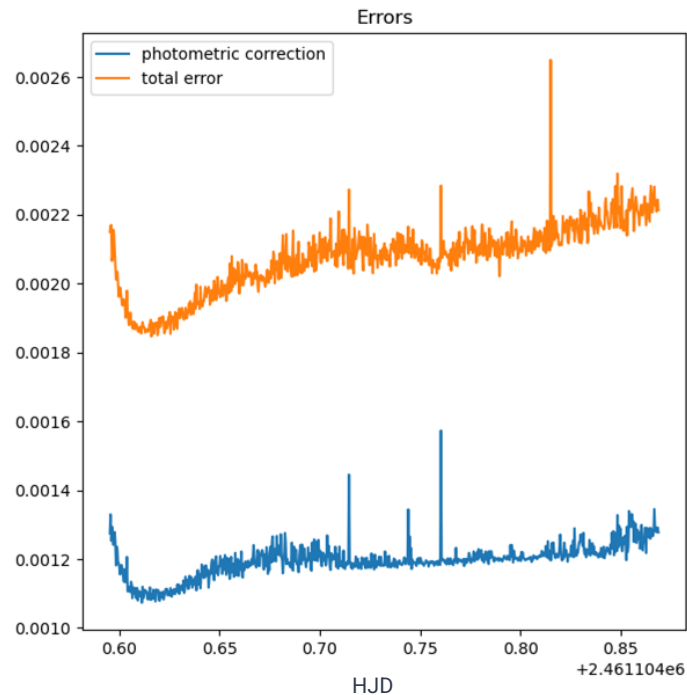
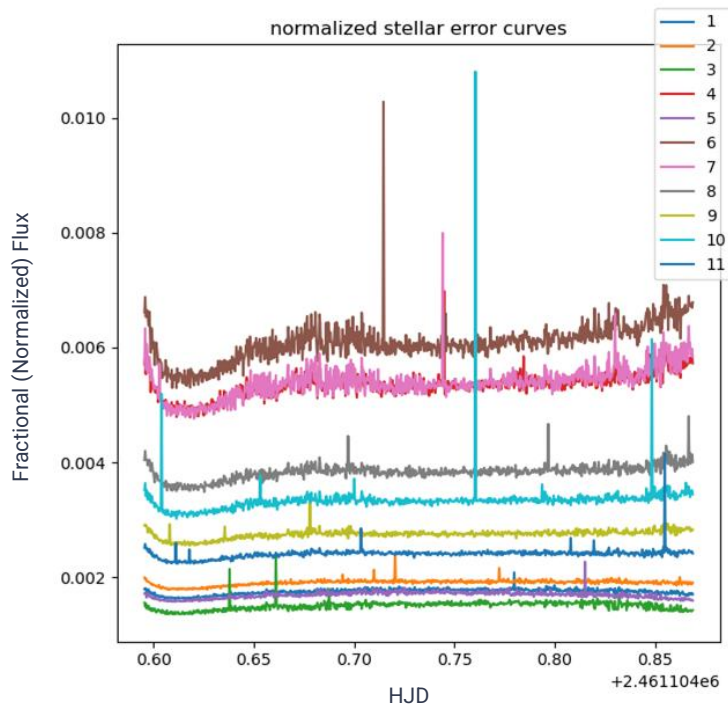


Can We Trust Our Heuristics?

We have a theoretical stellar pixel position for each circled star in our finder chart and average calibratory FWHMs from our observing run - we implement a recommended aperture radius of $2.25 \times \text{FWHM}$, but are these accurate?

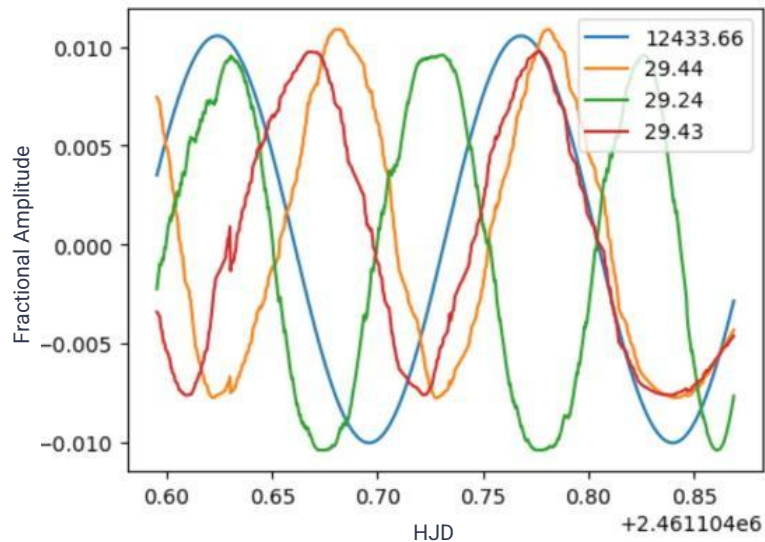
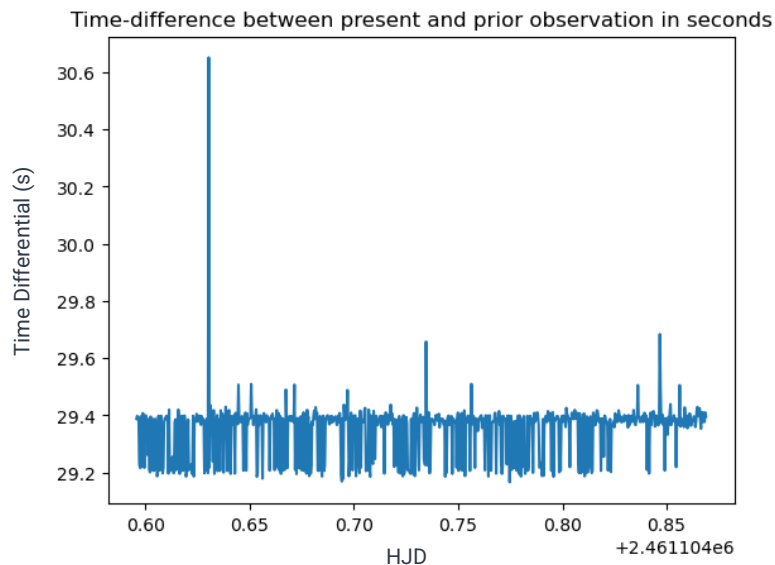


How Good is our Flux Resolution?

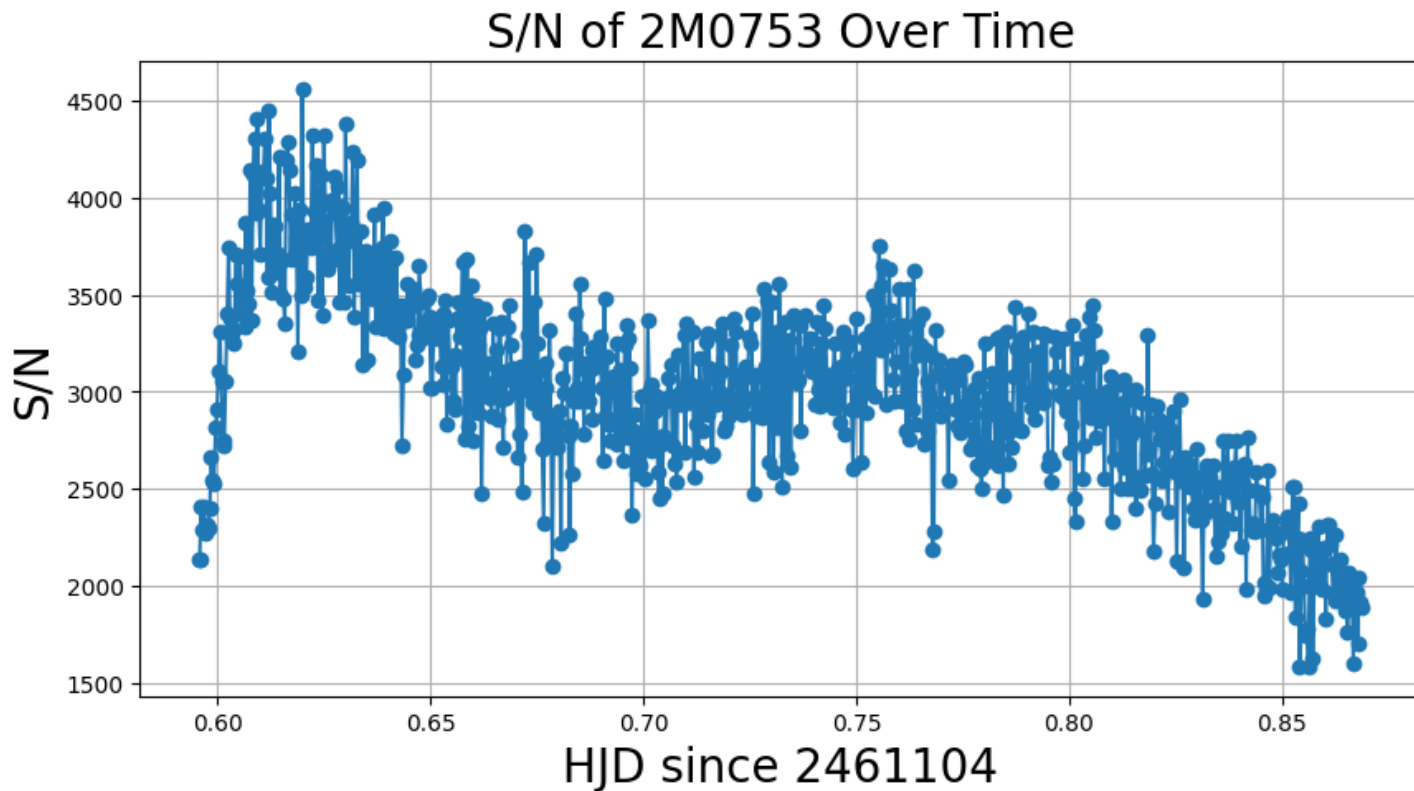


What is the Quality of our Frequency Data?

Irregularly sampled data can extract frequencies far in excess of the expected nyquist frequency for an equivalent Fourier transform (up to the LCM of every spacing between every data point), but has worse-defined errors and confidence. Our data is likely pseudo-periodic (in terms of aliasing and accessible frequency space constraints)

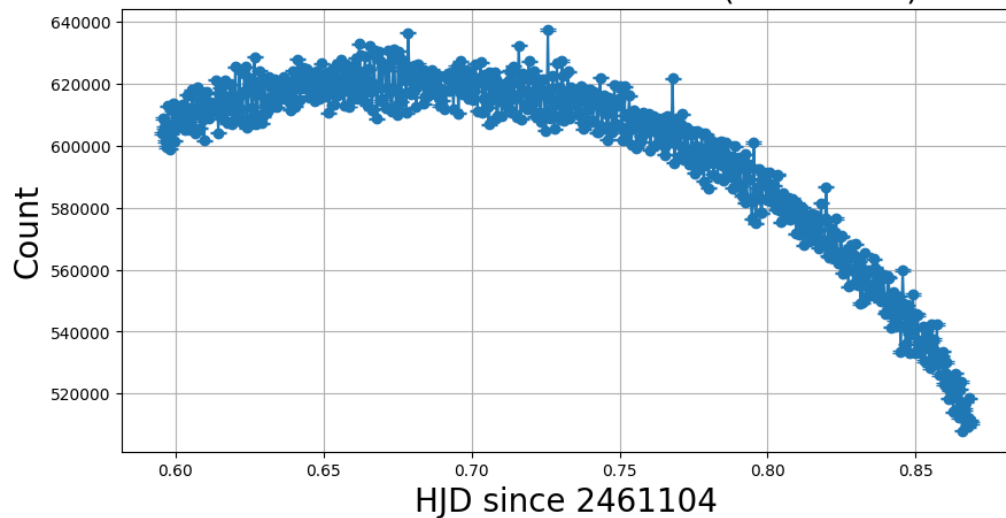


S/N of Target Over Time

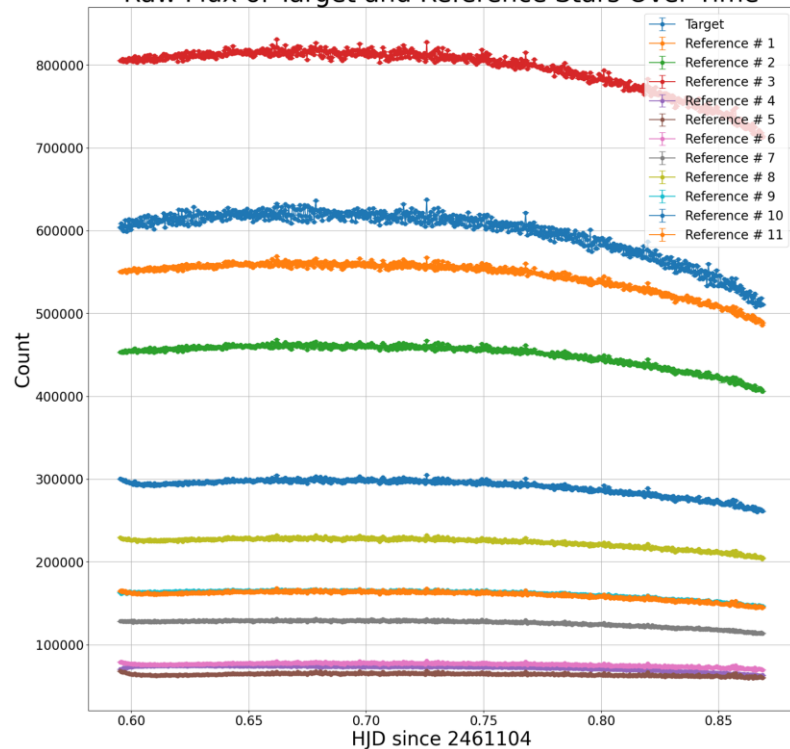


Raw Light Curves

Raw Flux of 2M0753 Over Time (with errors)

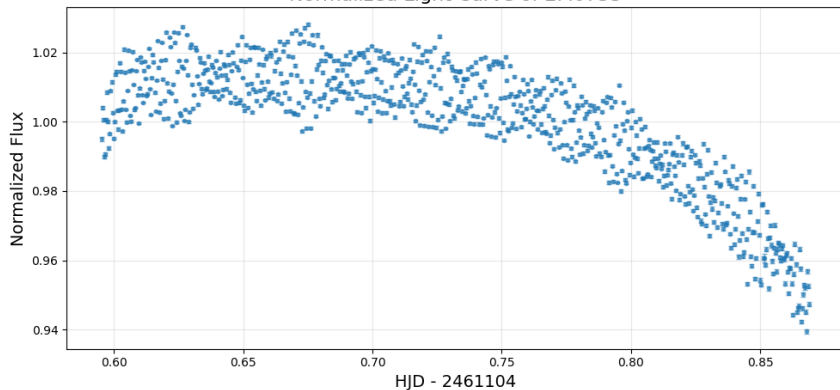


Raw Flux of Target and Reference Stars Over Time

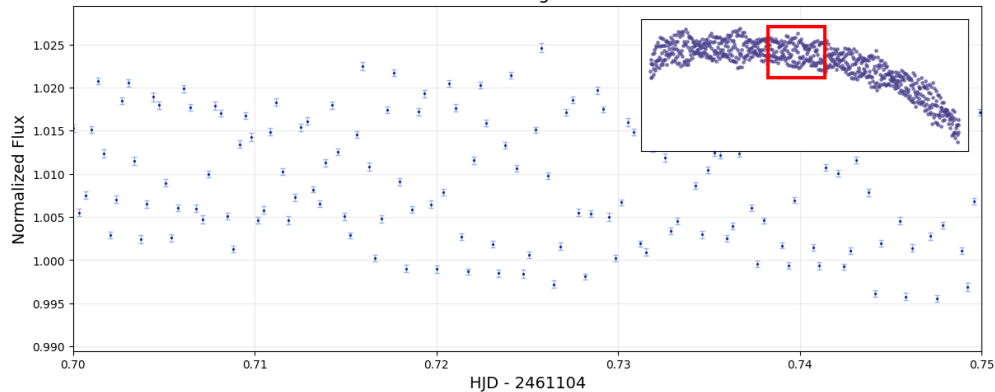


Normalized Light Curve

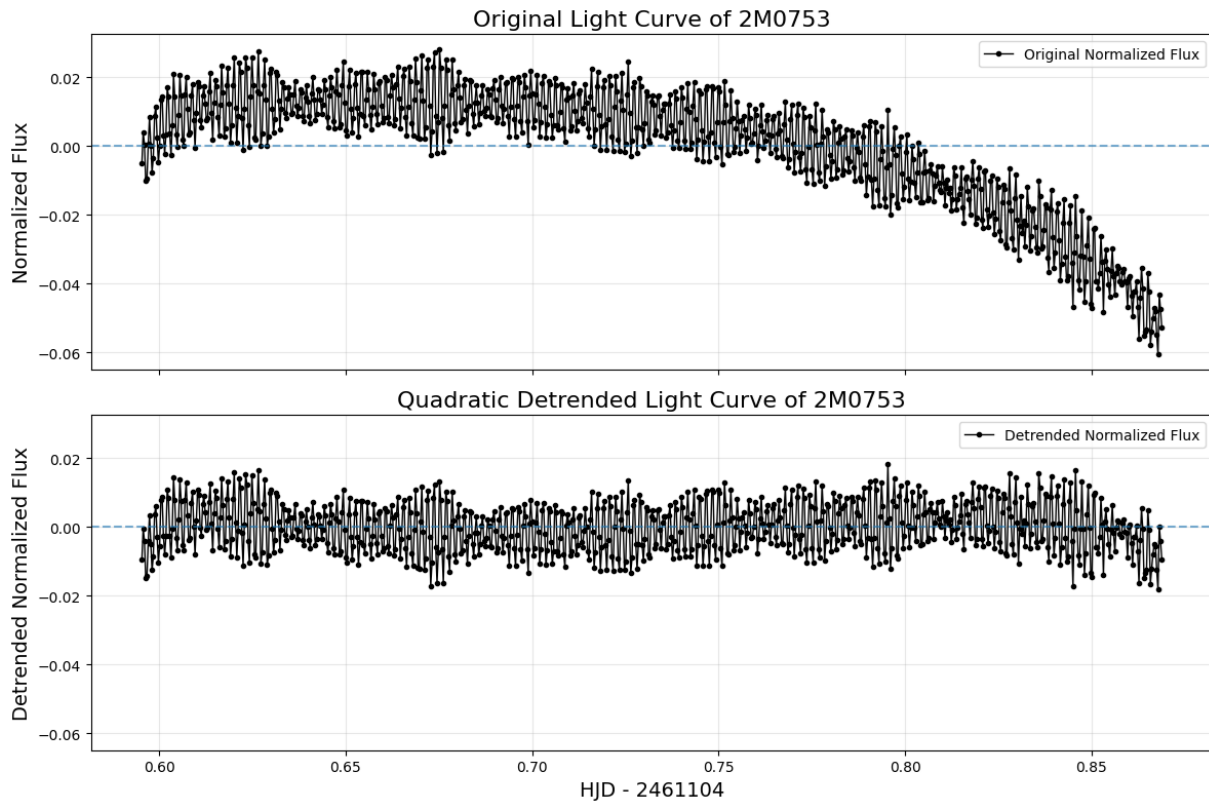
Normalized Light Curve of 2M0753



Zoomed Normalized Light Curve with Inset



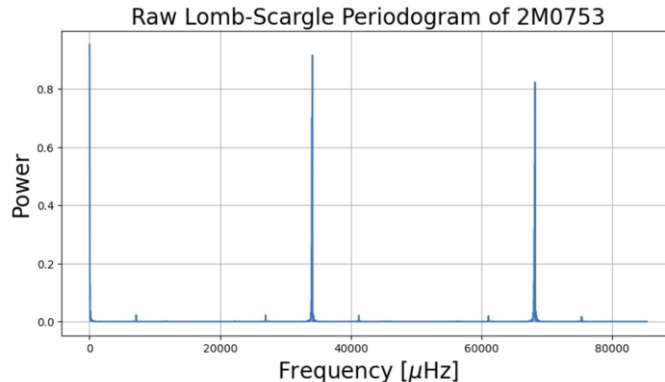
(Detrended) Normalized Light Curve



Periodograms

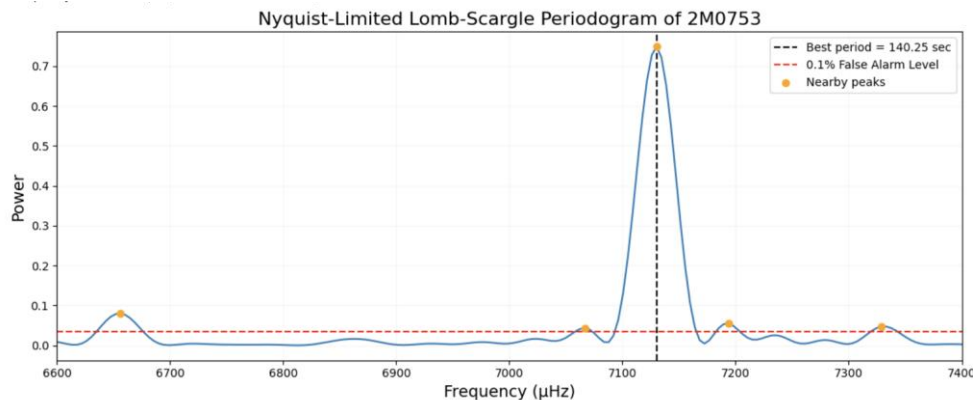
Raw Periodogram

- Many peaks, including above cut-off frequency
- High-frequency signals are not physically reliable
- Strong 1 cycle/day peak is leftover trend



Nyquist-Limited and Detrended

- Restricted to physically meaningful frequencies
- Periodogram becomes much cleaner
- Clear dominant peak at $\sim 7130 \mu\text{Hz}$ (~ 140 s)



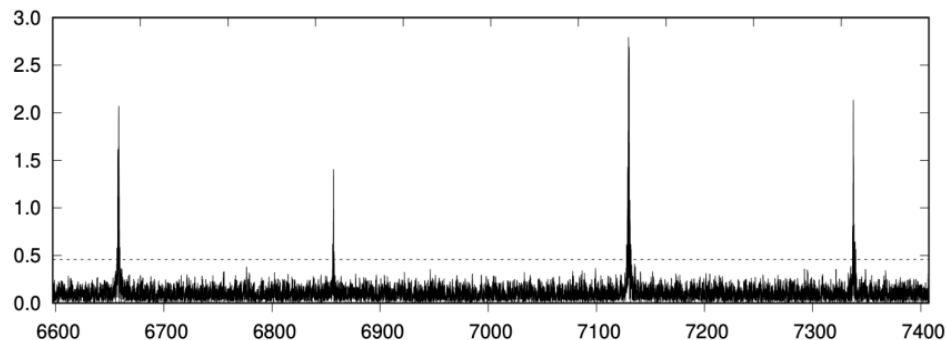
Periodograms Continued

Comparison to Paper:

- Paper finds multiple modes (135–150 s)
- We detect strong peak at ~140 s
- Smaller nearby peaks → likely unresolved modes

Sanity check data split:

- Two mini periodograms
 - First half main peak: 140.31 s
 - Second half main peak: 140.23 s



(Periodogram from “Short-period pulsating hot subdwarf stars observed by TESS”, Baran et. al)

Thank you!

...and May the fourth be with you.